🔐 Day 19 – Indicators of Compromise (IOC) Deep Dive

🎯 Objective

Understand how SOC analysts identify, analyze, and use IOCs to detect cyber threats.

📌 1. What is an IOC?

An **Indicator of Compromise (IOC)** is a piece of evidence that suggests a system, network, or device **may have been attacked or compromised**.

👉 It acts as a **warning signal** for security teams.

🧩 2. Types of IOCs

Common IOC categories used in SOC:

- 🌐 **IP Address** – Suspicious or malicious IP (e.g., attacker server)
- 🌍 **Domain Name** – Fake or phishing domains
- 🔗 **URL** – Malicious links used in attacks
- 📧 **Email Address** – Phishing or spoofed emails
- 🧬 **File Hash** – Unique fingerprint of malicious files (MD5, SHA256)

🧪 3. Example IOCs

- IP Address: 185.23.11.5
- Domain: malicious-login.com
- File Hash: abc123

👉 These are used to **identify known threats**.

🛡️ 4. How SOC Uses IOCs

SOC analysts use IOCs to:

- 🔍 **Search logs** for suspicious activity
- 🚨 **Detect attacks** in real-time
- ⛔ **Block malicious entities** (IP/domain/hash)
- 📊 **Investigate incidents**

⚙️ 5. Detection Logic (Basic Rule)

IF IP matches IOC
THEN generate alert

👉 This is how SIEM tools trigger alerts based on known threats.

⚠️ 6. Limitations of IOCs

- 🔄 **Short lifespan** – Attackers frequently change IOCs
- 🧠 **Only detects known threats** (not zero-day attacks)
- 🎯 Can be **easily bypassed** by advanced attackers

📝 Assignment Answers (Quick Help)

1. Define IOC:
An IOC is evidence that indicates a system may be compromised.

2. Examples:
IP address, domain, file hash, email, URL.

3. Limitations:
IOCs change quickly and only detect known threats.